Statistical Methods

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Outline I

1 Standard Distributions

Standard Discrete Distributions: The Discrete Uniform

We begin our exploration of standard distributions with the most fundamental concept of fairness: a system where every outcome in a finite set is equally likely.

The Discrete Uniform Distribution

Let a random variable X take on any integer value in a contiguous range from a to b inclusive.

- The total number of possible outcomes is $N = b - a + 1$.
- Because no outcome is favored, the probability is distributed evenly across the entire support.

Probability Mass Function (PMF)

We write $X \sim \text{DiscreteUniform}(a, b)$. The PMF is:

$$P(X = k) = \frac{1}{N} = \frac{1}{b - a + 1}, \quad \text{for } k = a, a + 1, \ldots, b$$

Discrete Uniform: Expectation and Variance

1. Expectation (The Midpoint)

Using the definition of expectation and factoring out the constant probability:

$$E[X] = \sum_{k=a}^{b} k \frac{1}{N} = \frac{1}{N} \sum_{k=a}^{b} k$$

The sum of this arithmetic sequence is the average of the first and last terms multiplied by the total number of terms (N):

$$E[X] = \frac{1}{N} \left(N \frac{a+b}{2} \right) = \frac{a+b}{2}$$

Discrete Uniform Variance

Calculating the sum of squares directly from a to b is messy. Instead, we define a shifted variable $Y = X - a + 1$.
Y now takes values $1, 2, \ldots, N$. Because shifting a distribution does not change its spread, $Var(X) = Var(Y)$.

1. Find the Second Moment $E[Y^2]$

Using the standard formula for the sum of consecutive squares:

$$E[Y^2] = \sum_{k=1}^{N} k^2 \frac{1}{N} = \frac{1}{N}\left[\frac{N(N+1)(2N+1)}{6}\right] = \frac{(N+1)(2N+1)}{6}$$

2. Calculate Final Variance

$$Var(Y) = \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2}\right)^2$$

$$\implies Var(Y) = \frac{N+1}{2}\left[\frac{N-1}{6}\right] = \frac{N^2-1}{12} \quad \blacksquare$$

The Continuous Uniform Distribution

We begin with the continuous equivalent of perfect fairness.

The Definition

A random variable $X \sim \text{Uniform}(a, b)$ is equally likely to take any real value in the continuous interval $[a, b]$. Because the density is evenly spread out, the PDF is a perfectly flat, horizontal line.

The PDF Formula

To ensure the total area under the curve equals exactly 1 (a rectangle of width $b - a$), the height must be $\frac{1}{b-a}$:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Continuous Uniform: Expected Value

Let $X \sim \text{Uniform}(a, b)$ with PDF $f(x) = \frac{1}{b-a}$. We find the expected value via integration:

$$\begin{aligned} E[X] &= \int_a^b x f(x) dx \\ &= \int_a^b x \left(\frac{1}{b-a} \right) dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \left(\frac{b^2 - a^2}{2} \right) \end{aligned}$$

Factoring the difference of squares in the numerator, $b^2 - a^2 = (b-a)(b+a)$:

$$E[X] = \frac{1}{b-a} \frac{(b-a)(b+a)}{2} = \frac{a+b}{2} \quad \blacksquare$$

Continuous Uniform: Variance

First, we calculate the second moment, $E[X^2]$:

$$E[X^2] = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$
$$= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2+ab+b^2)}{3(b-a)} = \frac{a^2+ab+b^2}{3}$$

Now, apply the variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$Var(X) = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2$$
$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$

Finding a common denominator of 12:

$$Var(X) = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$$
$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12} \quad \blacksquare$$

Generating Random Numbers

Computers are inherently deterministic, meaning they cannot generate truly random numbers. Instead, we rely on algorithms called **Pseudo-Random Number Generators (PRNGs)** to create sequences $U_1, U_2, \ldots$ that behave like standard uniforms, Uniform$(0, 1)$.

What must a PRNG ensure?

To be mathematically valid for simulation, the algorithm must guarantee:

- **Uniformity:** The numbers must be spread evenly across $[0, 1]$. No sub-interval should be favored.
- **Independence:** Successive numbers must lack correlation. Knowing U_n should give you no predictable information about U_{n+1}.
- **Long Period:** Because the system has finite memory and it is determeinistic, the sequence will eventually repeat. The length of this cycle (the "period") must be massive to avoid repeating patterns during a simulation.
- **Efficiency:** It must be computationally fast.

A Simple Algorithm: The Linear Congruential Generator

One of the oldest and most fundamental PRNG algorithms is the **Linear Congruential Generator (LCG)**. It generates a sequence of pseudo-random integers $X_0, X_1, X_2, \ldots$ using a simple recursive formula:

$$X_{n+1} = (aX_n + c) \pmod{m}$$

Where the algorithm is defined by four integer parameters:

- $m > 0$: the **modulus** (which dictates the maximum possible period)
- $a \in \{1, \ldots, m-1\}$: the **multiplier**
- $c \in \{0, \ldots, m-1\}$: the **increment**
- $X_0 \in \{0, \ldots, m-1\}$: the **seed** (the initial starting state)

To obtain the standard uniform output $U \sim \text{Uniform}(0, 1)$, we scale the result:

$$U_{n+1} = \frac{X_{n+1}}{m}$$

LCG: A "Toy" Example

Let's build a miniature LCG to see the algorithm in action.

The Setup

Let Modulus $m = 9$, Multiplier $a = 2$, Increment $c = 0$, and Seed $X_0 = 1$. The recurrence relation is: $X_{n+1} = (2X_n) \pmod 9$

- $X_1 = (2 \times 1) \pmod 9 = 2 \implies \mathbf{U_1 = 2/9 \approx 0.222}$
- $X_2 = (2 \times 2) \pmod 9 = 4 \implies \mathbf{U_2 = 4/9 \approx 0.444}$
- $X_3 = (2 \times 4) \pmod 9 = 8 \implies \mathbf{U_3 = 8/9 \approx 0.889}$
- $X_4 = (2 \times 8) \pmod 9 = 7 \implies \mathbf{U_4 = 7/9 \approx 0.778}$
- $X_5 = (2 \times 7) \pmod 9 = 5 \implies \mathbf{U_5 = 5/9 \approx 0.556}$
- $X_6 = (2 \times 5) \pmod 9 = 1 \implies \mathbf{U_6 = 1/9 \approx 0.111}$

Note: Modern algorithms use highly complex bitwise operations and massive prime moduli to achieve periods like $2^{19937} - 1$, satisfying the strict criteria from our first slide.

Simulating General Uniform Distributions

How do we get from $U \sim \text{Uniform}(0, 1)$ to other uniform distributions? We use simple linear transformations.

1. Continuous Uniform on $[a, b]$

To generate $X \sim \text{Uniform}(a, b)$, we scale U by the width of our target interval $(b - a)$, and then shift it by the starting point a:

$$X = a + (b - a)U$$

2. Discrete Uniform on $\{a, a + 1, \ldots, b\}$

To generate a random integer uniformly, we first create a continuous uniform variable over a slightly wider range $[a, b + 1)$, and then apply the floor function $\lfloor \cdot \rfloor$ to round down to the nearest integer:

$$X = \lfloor a + (b - a + 1)U \rfloor$$

Notice that the multiplier $(b - a + 1)$ is exactly N, the total number of distinct outcomes.

Generating from any CDF: The Inverse Transform Method

What if we want to simulate a random variable X from a distribution with a known Cumulative Distribution Function (CDF), $F(x)$?

The Generalized Inverse Transform Theorem

A valid CDF is non-decreasing but may have flat regions where a standard inverse is undefined. Instead, we use the **generalized (left) inverse**:

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$$

Let $U \sim \text{Uniform}(0, 1)$. If we define a new random variable as:

$$X = F^{-1}(U)$$

Then X is distributed exactly according to $F(x)$.

Generating from any CDF: The Inverse Transform Method

Proof: We want to show that the CDF of our new variable X is $F(x)$:

$$P(X \leq x) = P(F^{-1}(U) \leq x)$$

Because $F(x)$ is non-decreasing and right-continuous, the generalized inverse satisfies the fundamental equivalence $F^{-1}(u) \leq x \iff u \leq F(x)$. Therefore, we can rewrite the probability:

$$P(F^{-1}(U) \leq x) = P(U \leq F(x))$$

Since $U \sim \text{Uniform}(0, 1)$, we know $P(U \leq c) = c$ for any constant $c \in [0, 1]$. Because a valid CDF guarantees $0 \leq F(x) \leq 1$, it follows that:

$$P(U \leq F(x)) = F(x) \quad \blacksquare$$

Inverse Transform Method: Example

Example: The Exponential Distribution

Suppose X has CDF: $F(x) = 1 - e^{-\lambda x}, \quad \text{for } x \geq 0$

1. Set a standard uniform variable U equal to the CDF:

$$U = 1 - e^{-\lambda X}$$

2. Solve algebraically for X to find the inverse $F^{-1}(U)$:

$$e^{-\lambda X} = 1 - U \implies -\lambda X = \ln(1 - U) \implies X = -\frac{1}{\lambda} \ln(1 - U)$$

3. Because $U \sim \text{Uniform}(0, 1)$, the quantity $(1 - U)$ is also exactly distributed as Uniform$(0, 1)$. For computational efficiency, we often simplify the algorithm to:

$$X = -\frac{1}{\lambda} \ln(U)$$

Inverse Transform Method: Discrete Example

Example: A Generic Discrete Distribution

Suppose X takes positive integer values $i = 1, 2, 3, \ldots$ with probabilities $p_i = P(X = i)$. The CDF is a step function:

$$F(k) = P(X \leq k) = \sum_{i=1}^{k} p_i$$

Because the CDF is a step function, we rely entirely on the generalized inverse $F^{-1}(U) = \inf\{x : F(x) \geq U\}$. This essentially finds the first "step" of the CDF that reaches or exceeds U.

1. Generate a standard uniform variable $U \sim \text{Uniform}(0, 1)$.
2. Accumulate the probabilities $p_1, p_2, \ldots$ until the sum is greater than or equal to U. In other words, find the smallest integer k such that:

$$U \leq \sum_{i=1}^{k} p_i$$

Inverse Transform Method: Discrete Example

A Concrete Discrete Distribution

Suppose X takes three values with the following probabilities (p_i) and CDF values $F(x) = P(X \leq x)$:

- $P(X = 1) = 0.2 \implies F(1) = 0.2$
- $P(X = 2) = 0.5 \implies F(2) = 0.2 + 0.5 = 0.7$
- $P(X = 3) = 0.3 \implies F(3) = 0.7 + 0.3 = 1.0$

Finding the Generalized Inverse $F^{-1}(u)$

Recall $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. To find the inverse, we see where a given probability $u \in (0, 1]$ lands in our CDF steps:

$$F^{-1}(u) = \begin{cases} 1 & \text{if } 0 < u \leq 0.2 \\ 2 & \text{if } 0.2 < u \leq 0.7 \\ 3 & \text{if } 0.7 < u \leq 1.0 \end{cases}$$

The Bernoulli Trial

The Bernoulli Trial

A Bernoulli trial is a random experiment with only two possible outcomes, typically labeled as "Success" (1) and "Failure" (0).

- Let p be the probability of success, where $0 \leq p \leq 1$.
- The probability of failure is therefore $1 - p$ (often denoted as q).

Examples:

- Flipping a biased coin (Heads $= 1$, Tails $= 0$).
- Testing a manufactured component (Pass $= 1$, Defective $= 0$).
- A bit transmitted over a noisy channel (Received correctly $= 1$, Error $= 0$).

The Bernoulli Distribution

If a random variable X represents the outcome of a Bernoulli trial, we write $X \sim \text{Bernoulli}(p)$.
The PMF can be compactly written for $x \in \{0, 1\}$ as:

$$P(X = x) = p^x(1 - p)^{1-x}$$

Using our fundamental definitions of expectation and variance:

- **Expectation:**

$$E[X] = \sum_{x \in \{0,1\}} x \cdot P(X = x) = (1 \cdot p) + (0 \cdot (1 - p)) = p$$

- **Second Moment:**

$$E[X^2] = (1^2 \cdot p) + (0^2 \cdot (1 - p)) = p$$

- **Variance:**

$$Var(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

Standard Discrete Models: The Binomial Distribution

What happens if we repeat a Bernoulli trial multiple times?

Definition via Summation

Let $X_1, X_2, \ldots, X_n$ be n **independent** random variables, where each $X_i \sim \text{Bernoulli}(p)$.
We define a new random variable Y as their sum:

$$Y = \sum_{i=1}^{n} X_i$$

Y represents the total number of "Successes" in n independent trials. We say $Y \sim \text{Binomial}(n, p)$.

Examples:

- The number of Heads in 10 coin flips.
- The number of defective items in a batch of 50.

Binomial Probability Mass Function

To find the PMF of Y, we want the probability of getting exactly k successes (where $k \in \{0, 1, \ldots, n\}$).

Deriving the Probability

1. **Specific Sequence:** The probability of getting k successes followed exactly by $n - k$ failures is $p^k(1-p)^{n-k}$.
2. **Counting Combinations:** There are many ways to arrange k successes among n trials. The number of such unique sequences is given by the binomial coefficient $\binom{n}{k}$.

Multiplying these together yields the PMF:

$$P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Binomial Mean and Variance

We could calculate the expectation by solving $\sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}$, which requires complex algebra.
Instead, we use the linear properties we proved earlier!

1. Expectation (Linearity)

Because expectation is a linear operator:

$$E[Y] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p = np$$

2. Variance (Independence)

Because the trials X_i are **independent**, the covariance terms vanish, allowing us to distribute the variance operator:

$$Var(Y) = Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) = \sum_{i=1}^{n} p(1-p) = np(1-p)$$

Binomial Expectation:

Let $Y \sim \text{Binomial}(n, p)$ and $q = 1 - p$. We calculate $E[Y]$ directly:

$$E[Y] = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^{n} k \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

Cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[Y] = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k}$$

Factor out np to reconstruct a new binomial coefficient:

$$E[Y] = np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{(n-1)-(k-1)}$$

Let $j = k - 1$. By the Binomial Theorem, the remaining sum is $(p+q)^{n-1} = 1$:

$$E[Y] = np \cdot 1 = np \quad \blacksquare$$

Binomial Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[Y(Y-1)]$:

$$E[Y(Y-1)] = \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k q^{n-k}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting at $k = 2$, cancel $k(k-1)$ with $k!$:

$$E[Y(Y-1)] = \sum_{k=2}^{n} \frac{n!}{(k-2)!(n-k)!} p^k q^{n-k}$$

Factor out $n(n-1)p^2$ to rebuild the combinations:

$$E[Y(Y-1)] = n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k}$$

Just as before, the remaining sum is exactly $(p+q)^{n-2} = 1$. Thus:

$$E[Y(Y-1)] = n(n-1)p^2$$

Binomial Variance:

We just found $E[Y(Y-1)] = n(n-1)p^2$ and we know $E[Y] = np$.

1. Find the Second Moment $E[Y^2]$

Expand the factorial moment: $E[Y(Y-1)] = E[Y^2 - Y] = E[Y^2] - E[Y]$.

Rearranging gives:

$$\begin{aligned} E[Y^2] &= E[Y(Y-1)] + E[Y] \\ &= n(n-1)p^2 + np \\ &= n^2p^2 - np^2 + np \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $Var(Y) = E[Y^2] - (E[Y])^2$:

$$\begin{aligned} Var(Y) &= (n^2p^2 - np^2 + np) - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np - np^2 = np(1-p) \quad \blacksquare \end{aligned}$$

The Sample Proportion ($\hat{p}$)

Instead of looking at the total successes Y, let's examine the **average number of successes** per trial, known as the sample proportion: $\hat{p} = \frac{Y}{n}$.

1. Expectation

$$E\left[\frac{Y}{n}\right] = \frac{1}{n}E[Y] = \frac{1}{n}(np) = p$$

The expected proportion of successes is exactly the probability of a single success.

2. Variance

Recall that pulling a constant out of a variance operator squares the constant:

$$Var\left(\frac{Y}{n}\right) = \frac{1}{n^2}Var(Y) = \frac{1}{n^2}np(1-p) = \frac{p(1-p)}{n}$$

The Sample Proportion: As $n \to \infty$

We just established that the variance of the sample proportion is inversely proportional to the number of trials:

$$Var(\hat{p}) = \frac{p(1-p)}{n}$$

The Infinite Limit

What happens mathematically if we repeat the trial an infinite number of times?

$$\lim_{n \to \infty} Var(\hat{p}) = \lim_{n \to \infty} \frac{p(1-p)}{n} = 0$$

Physical Implication: As n grows indefinitely, the sample proportion $\hat{p}$ stops fluctuating and collapses precisely onto the true, underlying probability p. This validates the frequentist definition of probability.

Sums of Independent Random Variables

Let X and Y be two **independent** random variables defined on the same probability space. We are interested in finding the probability distribution of their sum:

$$Z = X + Y$$

To determine the distribution of Z, we analyze all possible pairs (x, y) such that $x + y = z$. Because Y is constrained by $Y = z - X$, independence allows us to determine the distribution using the mathematical operation of **convolution**.

The Discrete Case: PMF Convolution

If X and Y are discrete independent random variables with PMFs $p_X(x)$ and $p_Y(y)$, the PMF of Z is obtained by summing over all valid disjoint supports:

$$p_Z(z) = P(X + Y = z) = \sum_x p_X(x) p_Y(z - x)$$

This operation is frequently denoted as $p_Z = p_X * p_Y$.

Convolution of Continuous Random Variables

The Continuous Case: PDF Convolution

If X and Y are continuous independent random variables characterized by the PDFs $f_X(x)$ and $f_Y(y)$, the density of their sum Z is evaluated via integration:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$$

Property: Commutativity

Because algebraic addition is commutative ($X + Y = Y + X$), the convolution operator is symmetric. Thus, the order of the functions under the integral can be inverted:

$$\int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y)\, dy$$

Example: Discrete Support (Geometric Sum)

Problem Setup

Let X and Y be independent Geometric random variables representing the number of trials until the first success, both with success probability p. Their PMF for $k \in \{1, 2, 3, \dots\}$ is:

$$p_X(k) = p_Y(k) = (1-p)^{k-1}p$$

Find the PMF of $Z = X + Y$.

Applying the Convolution Formula

Since the minimum value for both X and Y is 1, the minimum value for Z is $1 + 1 = 2$.

$$p_Z(z) = \sum_x p_X(x)p_Y(z-x)$$

The lower limit for x is 1. For $p_Y(z-x)$ to be non-zero, we also need $z - x \geq 1 \implies x \leq z - 1$.

$$\sum_{x=1}^{z-1}$$

Example: Geometric Sum (Solution)

Simplifying the Summation

We combine the terms with base $(1-p)$ and group the p terms:

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{x-1+z-x-1}$$

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{z-2}$$

Final Result

Since the term inside the sum does not depend on x, we are simply adding a constant to itself $z-1$ times:

$$p_Z(z) = (z-1)p^2(1-p)^{z-2}, \quad \text{for } z = 2, 3, 4, \ldots$$

Note: This is exactly the Negative Binomial distribution $NB(2, p)$.

The "Rare Event" Limit

What happens to a Binomial distribution when the number of trials n becomes extremely large, but the probability of success p is extremely small?

The Setup

Consider modeling rare events in a massive population, such as:

- The number of typos in a 500-page book.
- The number of radioactive decays in a millisecond.

Let us take the limit as $n \to \infty$ and $p \to 0$. To keep the mathematics meaningful, we hold their product constant: let $np = \lambda$.
Here, λ represents the **average rate** of successes.

Deriving the Poisson PMF: Setup

Let $Y \sim \text{Binomial}(n, p)$. We substitute $p = \frac{\lambda}{n}$ into the Binomial Probability Mass Function and take the limit as $n \to \infty$:

$$\lim_{n\to\infty} P(Y = k) = \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \lim_{n\to\infty} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

We can group the constants (with respect to n) and pull them to the front:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \left[\frac{n!}{(n-k)!} \cdot \frac{1}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} \right]$$

Deriving the Poisson PMF: Evaluation

Let's break the expression inside the limit into three distinct components:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \underbrace{\left[\frac{n(n-1)\dots(n-k+1)}{n^k}\right]}_{\text{Part A}} \underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\text{Part B}} \underbrace{\left(1-\frac{\lambda}{n}\right)^{-k}}_{\text{Part C}}$$

1. **Part A:** There are exactly k terms in the numerator polynomial and n^k in the denominator. As $n \to \infty$, the highest degree terms dominate, and this fraction approaches 1.
2. **Part B:** From standard calculus, the limit definition of the exponential function dictates that $\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda}$.
3. **Part C:** For any finite k, as $n \to \infty$, the term inside the parentheses approaches 1. Thus, $1^{-k} = 1$.

The Poisson Distribution

Multiplying our evaluated limits together $(1 \cdot e^{-\lambda} \cdot 1)$, we arrive at the PMF for a new, fundamental distribution.

The Poisson PMF

A random variable X follows a Poisson distribution with rate parameter $\lambda > 0$, denoted $X \sim \text{Poisson}(\lambda)$, if:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \ldots$$

Mean and Variance of the Poisson: Intuitive

Because we derived this by holding $np = \lambda$ constant:

- **Mean:** $E[X] = \lim_{n\to\infty}(np) = \lambda$
- **Variance:** $Var(X) = \lim_{n\to\infty} np(1-p) = \lambda(1-0) = \lambda$

Poisson Expectation: Direct Derivation

Let $X \sim \text{Poisson}(\lambda)$. By definition, the expected value is:

$$E[X] = \sum_{k=0}^{\infty} kP(X = k) = \sum_{k=0}^{\infty} k\frac{\lambda^k e^{-\lambda}}{k!}$$

The term for $k = 0$ is exactly zero, so we can start the sum at $k = 1$. We cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[X] = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!}$$

We factor out $\lambda e^{-\lambda}$ to shift the index. Let $j = k - 1$:

$$E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion for $e^{\lambda} = \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$:

$$E[X] = \lambda e^{-\lambda}(e^{\lambda}) = \lambda \quad \blacksquare$$

Poisson Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k e^{-\lambda}}{k!}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting the sum at $k = 2$, we cancel $k(k-1)$ with the $k!$ in the denominator:

$$E[X(X-1)] = \sum_{k=2}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!}$$

Factor out $\lambda^2 e^{-\lambda}$ and let $j = k - 2$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Once again, the infinite sum resolves exactly to e^{λ}:

$$E[X(X-1)] = \lambda^2 e^{-\lambda}(e^{\lambda}) = \lambda^2$$

Poisson Variance: Direct Derivation

We now have the two necessary components: $E[X(X-1)] = \lambda^2$ and $E[X] = \lambda$.

1. Find the Second Moment $E[X^2]$

Expand the factorial moment: $E[X(X-1)] = E[X^2] - E[X]$.

Rearranging gives us:

$$\begin{aligned} E[X^2] &= E[X(X-1)] + E[X] \\ &= \lambda^2 + \lambda \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$\begin{aligned} Var(X) &= (\lambda^2 + \lambda) - (\lambda)^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \quad \blacksquare \end{aligned}$$

Waiting Time: The Geometric Distribution

We return to our sequence of independent Bernoulli trials (where $P(\text{Success}) = p$ and $P(\text{Failure}) = q = 1 - p$).
Instead of counting the *number* of successes in a fixed time, we now ask:
How long must we wait until the very first success?

Defining the Random Variable

Let X be the number of trials needed to get the first success.

- The only way for $X = k$ to occur is to have exactly $k - 1$ failures followed immediately by 1 success.
- Because the trials are independent, we multiply their probabilities.

The Geometric PMF

We say $X \sim \text{Geometric}(p)$. Its Probability Mass Function is:

$$P(X = k) = q^{k-1}p, \quad \text{for } k = 1, 2, 3, \ldots$$

Geometric Expectation:

By definition, $E[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} kq^{k-1}p$.
We can pull the constant p outside the sum:

$$E[X] = p\sum_{k=1}^{\infty} kq^{k-1}$$

To evaluate this infinite series, we recall the standard geometric series for $|q| < 1$:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

If we take the derivative of both sides with respect to q, the k drops down:

$$\frac{d}{dq}\sum_{k=0}^{\infty} q^k = \sum_{k=1}^{\infty} kq^{k-1} = \frac{d}{dq}(1-q)^{-1} = \frac{1}{(1-q)^2}$$

Substituting this result back into our expectation formula:

$$E[X] = p\left[\frac{1}{(1-q)^2}\right] = p\left[\frac{1}{p^2}\right] = \frac{1}{p} \quad \blacksquare$$

Geometric Variance and Intuition

To find the variance, we would use the second derivative of the geometric series to find the second factorial moment $E[X(X-1)]$.

Moments Summary

For $X \sim \text{Geometric}(p)$:

- **Mean:** $E[X] = \frac{1}{p}$
- **Variance:** $Var(X) = \frac{1-p}{p^2}$

Physical Intuition

The expected value $E[X] = 1/p$ aligns perfectly with human intuition:

- If you flip a fair coin ($p = 1/2$), you expect to wait $1/(1/2) = 2$ flips to see the first Heads.
- If you roll a fair die looking for a six ($p = 1/6$), you expect to wait $1/(1/6) = 6$ rolls to see the first 6.

The Memoryless Property

A unique and defining characteristic of the Geometric distribution is that it is **memoryless**.

Formal Definition

For any integers $s > 0$ and $t > 0$:

$$P(X > s + t \mid X > s) = P(X > t)$$

The probability of waiting t additional trials for a success, given that we have already waited s trials, is exactly the same as the probability of waiting t trials from the very beginning.

The samples do not "remember" past failures. Suppose you are rolling a fair die waiting for a 6.

- You have already rolled 10 times and failed.
- The die is not "due" for a 6. The probability that it takes 3 *more* rolls is exactly the same as taking 3 rolls on a brand new die.

Proof of the Memoryless Property

Prerequisite: Check that $P(X > k) = q^k$. (To fail to get a success in the first k trials means getting k failures in a row).

Using the definition of conditional probability:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \text{ and } X > s)}{P(X > s)}$$

Since the event $\{X > s + t\}$ is a subset of the event $\{X > s\}$, their intersection is simply $\{X > s + t\}$. We substitute our tail probabilities:

$$\begin{aligned} P(X > s + t \mid X > s) &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{q^{s+t}}{q^s} \\ &= q^t \\ &= P(X > t) \quad \blacksquare \end{aligned}$$

Conclusion: The past history of failures cancels out.

Generalizing the Wait: Negative Binomial

We just looked at the waiting time for the first success. Now, we generalize: **How long must we wait until the r-th success?**

Defining the Random Variable

Let X be the number of trials needed to achieve exactly r successes.

- The Geometric distribution is simply a special case where $r = 1$.
- For $X = k$ to occur, the very last trial (trial k) must be a success.
- The previous $k - 1$ trials must contain exactly $r - 1$ successes.

The Negative Binomial PMF

Let's build the Probability Mass Function for $X \sim \text{NB}(r, p)$.

Deriving the Formula

- Probability of exactly $r-1$ successes in the first $k-1$ trials (Binomial): $\binom{k-1}{r-1} p^{r-1} q^{(k-1)-(r-1)}$
- Probability that the k-th trial is a success: p
- Multiplying these independent events gives us our PMF!

The PMF

$$P(X = k) = \binom{k-1}{r-1} p^r q^{k-r}$$

for $k = r, r+1, r+2, \ldots$

Negative Binomial Moments

We could use calculus tricks to find the expectation, but there is a much more elegant way using linearity of expectation!

Sum of Independent Geometrics

We can think of waiting for r successes as waiting for the first success, then waiting for the second, and so on.

$$X = Y_1 + Y_2 + \cdots + Y_r$$

where each $Y_i \sim \text{Geometric}(p)$.

Moments Summary

Because expectation is linear and the wait times are independent:

- **Mean:** $E[X] = r \cdot E[Y_i] = \frac{r}{p}$
- **Variance:** $Var(X) = r \cdot Var(Y_i) = \frac{r(1-p)}{p^2}$

The Fish Population Problem

We want to estimate the total number of fish in a lake, but we obviously cannot drain the lake to count them all. Instead, we use a statistical inference technique called **Capture-Recapture**.

The Setup

- **Step 1 (Tagging):** We capture a specific number of fish (let's call this K), tag them securely, and release them back into the water to mix with the general population.
- **Step 2 (Sampling):** After giving them time to disperse, we capture a new sample of n fish from the lake.

The Random Variable

Let X be the number of tagged fish that end up in our second sample. We want to use X to estimate the total, unknown population size, N.

The Hypergeometric PMF

Because we are pulling fish from a finite lake and not throwing them back between individual catches, the trials are **not independent**. The Binomial distribution assumes replacement; the **Hypergeometric** handles scenarios *without* replacement.

The Parameters

- N: Total population size (unknown)
- K: Total tagged fish in the lake
- n: Our sample size (number of draws)
- k: The observed number of tagged fish in our sample

To find the probability of catching exactly k tagged fish, we divide the number of successful combinations by the total possible combinations:

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

Estimating the Total Population

Using Expectation for Inference

The expected value of a Hypergeometric random variable relies on the proportion of "successes" in the overall population:

$$E[X] = n\frac{K}{N}$$

- We use our actual observed number of tagged fish, k, as our best estimate for the expected value $E[X]$.

Solving for N: By substituting k in place of the expectation, we can rearrange the algebra to estimate the total population size:

$$N \approx \frac{n \cdot K}{k}$$

From Discrete to Continuous

Until now, our random variables jumped between distinct, separate values (counts, dice rolls, iterations). Now, we enter the continuous realm, where a variable can take **any real number** within a range (like exact weight, continuous time, or distance).

The Probability Density Function (PDF)

Instead of a Probability Mass Function (PMF), continuous variables are described by a PDF, denoted as $f(x)$.

- $f(x)$ is **not** a probability itself. It is a density.
- Probabilities are calculated by finding the **area under the curve** over a specific interval.

Because a continuous range has infinitely many points, the probability of hitting one exact, infinitely precise number is always zero:

$$P(X = c) = \int_c^c f(x)dx = 0$$

Deriving the Exponential: The Setup

Assume we have a continuous random variable $X \geq 0$ that possesses the memoryless property:

$$P(X > s + t \mid X > s) = P(X > t) \quad \text{for all } s, t \geq 0$$

Using the definition of conditional probability:

$$\frac{P(X > s + t \text{ and } X > s)}{P(X > s)} = P(X > t)$$

Since $(X > s + t)$ is a strict subset of $(X > s)$, their intersection is just $(X > s + t)$:

$$\frac{P(X > s + t)}{P(X > s)} = P(X > t) \implies P(X > s + t) = P(X > s)P(X > t)$$

To make this easier to read, let us define the **tail probability** (or survival function) as $G(x) = P(X > x)$. We now have a strict functional equation:

$$G(s + t) = G(s)G(t)$$

Deriving the Exponential Distribution

We need to find a tail probability function $G(x)$ that satisfies our memoryless equation:

$$G(s+t) = G(s)G(t)$$

Finding the Function

The standard algebraic function that converts addition in its input to multiplication in its output is the exponential function:

$$e^{\lambda(s+t)} = e^{\lambda s} \cdot e^{\lambda t}$$

Because $G(x) = P(X > x)$ represents a probability, it has strict boundaries:

- It must start at $G(0) = 1$.
- It must strictly decrease as the waiting time x increases.

To make the function decay rather than grow infinitely, the exponent must be negative. Thus, for some rate parameter $\lambda > 0$:

$$G(x) = e^{-\lambda x}$$

The Exponential Distribution

We just proved that the tail probability must be $P(X > x) = e^{-\lambda x}$.

1. Find the CDF

The Cumulative Distribution Function, $F(x) = P(X \leq x)$, is simply the complement of the tail probability:

$$F(x) = 1 - P(X > x) = 1 - e^{-\lambda x}$$

2. Find the PDF

By the Fundamental Theorem of Calculus, the Probability Density Function $f(x)$ is the derivative of the CDF:

$$\begin{aligned} f(x) &= \frac{d}{dx} F(x) \\ &= \frac{d}{dx} \left(1 - e^{-\lambda x}\right) \\ &= -e^{-\lambda x} \cdot (-\lambda) \\ &= \lambda e^{-\lambda x} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Mean Derivation

Let $X \sim \text{Exponential}(\lambda)$ with PDF $f(x) = \lambda e^{-\lambda x}$. By definition:

$$E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx$$

We use Integration by Parts (IBP): $\int u\, dv = uv - \int v\, du$.

- Let $u = x \implies du = dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned} E[X] &= \left[-xe^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) dx \\ &= (0 - 0) + \int_0^\infty e^{-\lambda x} dx \quad \left(\text{Note: } \lim_{x\to\infty} \frac{x}{e^{\lambda x}} = 0 \text{ by L'Hôpital's}\right) \\ &= \left[-\frac{1}{\lambda} e^{-\lambda x}\right]_0^\infty \\ &= 0 - \left(-\frac{1}{\lambda}\right) = \frac{1}{\lambda} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Second Moment

To find the variance, we first need $E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$. Using IBP again:

- Let $u = x^2 \implies du = 2x\,dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$E[X^2] = \left[-x^2 e^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) \cdot 2x\,dx$$
$$= 0 + 2\int_0^\infty x e^{-\lambda x} dx$$

Notice that the remaining integral is almost $E[X]$. We multiply and divide by λ:

$$E[X^2] = \frac{2}{\lambda} \int_0^\infty x \lambda e^{-\lambda x} dx$$
$$= \frac{2}{\lambda} \cdot E[X] = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2}$$

Exponential Distribution: Variance

Now we apply the variance formula:

$$Var(X) = E[X^2] - (E[X])^2$$

The Final Calculation

Substituting our derived moments:

$$\begin{aligned} Var(X) &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \\ &= \frac{1}{\lambda^2} \quad \blacksquare \end{aligned}$$

Summary for $X \sim \text{Exp}(\lambda)$

- **Mean:** $E[X] = 1/\lambda$
- **Standard Deviation:** $SD(X) = 1/\lambda$